

Informe de experimento: react_nogate

Generado el 2026-09-18 21:59 por benchmarks/experiment_report.py a partir de 1 report(s) del directorio react_nogate (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f  
↪ low-agent/benchmarks/results_validation_n10/gpt-5.6-sol/react_nogate
```

- Reports leídos: 1 (report.rescored.json usado en 0; el resto report.json)
- Modelos: openrouter:openai/gpt-5.6-sol
- Métodos: react_nogate
- Casos: case14
- Requests: generadas, N=10 por caso y seed, difficulties=todas (1 report)
- Items por dificultad: plain=3, parameterized=3, multistep=2, ambiguous=2
- System prompt: hash 7da5e2a37ec4 (10 filas)
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-18 04:15
- Costo total: 0.3268 USD (10 items; costo medio 0.03268 USD/item)
- Items totales: 10; ok=10; con error=0; tokens totales=144348
- Solved global: 70.0% (7/10)

Errores por tipo:

(ningún item con error)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo. **Esta reconstrucción es ilustrativa, no histórica:** usa el código de llm/prompts.py de HOY, así que si el texto del prompt cambió después de que esta corrida se generó, lo que se ve aquí no es lo que el modelo recibió. El registro histórico real es el hash por fila (system_prompt_hash, ver sección 1 / benchmarks/experiment_report.py::section_what_ran), fijado en el momento de la corrida y nunca recalculado.

Método react_nogate (system_prompt_hash real de esta fila: 7da5e2a37ec4)

System prompt (SYSTEM_PROMPT_EN) (reconstrucción con el código actual, no necesariamente histórica) — 1557 caracteres; architecture=react, gate=False, memory=False

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given, unless the request itself states a
  ↪ different indexing convention for that identifier (e.g. "(0-based)"), in which case convert it to the MATPOWER 1-based id before
  ↪ calling any tool.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
```

3. Violations, if any
4. Suggested next step

User prompt (primer item) = request_text −67 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

Load case14 and disconnect the line between bus ten and bus eleven.

3. Resultados agregados

Columnas del protocolo (medias por método × caso, del **scoreboard_per_case**), en los 4 grupos de **metricas_pfagent_definicion**: **utilidad de tarea** Form. (formulación exacta de las tool calls), V_MAE (p.u., solo sobre peticiones con formulación exacta −toda fila con herramientas que llega al solver con la formulación correcta da exactamente cero, ver benchmarks/scoring.py) y F_MAE (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status (convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (por respuesta, no por número: fracción de respuestas donde todo número es trazable a una salida de tool Y viene del solve posterior al último cambio de red) y SFR (fallos declarados con seguridad); **costo** Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Escalated y Wrong (silent). Escalated y Wrong (silent) junto con solved_automously_rate (no la columna Solved de arriba, que puede solaparse con Escalated −una abstención de verificación puede caer sobre un item cuyo cálculo de fondo sí era correcto) suman 100 %: dónde termina cada petición −resuelta sola, entregada a una persona, o mal contestada sin aviso; solo cuenta como Escalated una arquitectura con mecanismo de traspaso real, ver benchmarks.scoring.escalation_check) y \$ (costo total USD). Porcentajes en %.

Por método × caso

Method	Case	N	Form.	V_MAE	F_MAE	Solved	Solver status	B_mean	Faith.	SFR	Calls	Tok.	Claim	Escalated	Wrong (silent)	\$
react_nogate	case14	10	70.0 (7/10)	0	0.00	70.0 (7/10)	100.0 (10/10)	7.73e-07	90.0 (9/10)	−	3.2	14435−	0.0 (0/10)	30.0 (3/10)		0.3268

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	plain	parameterized	multistep	ambiguous
react_nogate	33.3 (1/3)	100.0 (3/3)	50.0 (1/2)	100.0 (2/2)

Solved por dificultad (% , items resueltos / items)

Method	plain	parameterized	multistep	ambiguous
react_nogate	33.3 (1/3)	100.0 (3/3)	50.0 (1/2)	100.0 (2/2)

Tipos de error de formulación (conteo de items)

Method	extra_arg	changes_result	extra_step	ok
react_nogate	2		1	7

Errores de ejecución por método (conteo de items con error)

(ningún item con error)

4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (claimed_success_on_failure, stale_state, abstained). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de **traces/** cuando existe.

Método react_nogate

Ejemplo: resuelto (solved=True)

```
request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 10, "to_bus": 11})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.782856812869876, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": -5930 chars más]
gate: PASS (reasons=-, mismatch=7.85e-09 MW)
--- Ronda 3 ---
LLM: 1. **Summary:** IEEE 14-bus case loaded successfully; line 10-11 was disconnected, and the AC power flow converged.

2. **Key numbers:**
- Total load: **253.846 MW**
- Total generation: **266.510 MW**
- Total losses: **12.665 MW**
- Voltage range: **1.010-1.090 p.u.**

3. **Violations:** ...[225 chars más]
--- Respuesta final ---
1. **Summary:** IEEE 14-bus case loaded successfully; line 10-11 was disconnected, and the AC power flow converged.

2. **Key numbers:**
- Total load: **253.846 MW**
- Total generation: **266.510 MW**
- Total losses: **12.665 MW**
- Voltage range: **1.010-1.090 p.u.**

3. **Violations:**
- Overvoltage at buses **1, 6, 7, 8, 11, and 12**
- Highest voltage: **1.090 p.u. at bus 8**
- No thermal violations

4. **Suggested next step:** Evaluate voltage-control remedial actions to reduce the overvoltages.
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=5430;
↳ cost_usd=0.0131; wall=12.5 s
```

Ejemplo: formulación fallida (formulation_exact=False)

```
request (case14-multistep-006-s0, case14, difficulty=multistep, seed=556019486): Load the 14-bus test case (case14), set the active load at
↳ bus 4 to 42.7 MW, disconnect the line between bus 3 and bus 4, reconnect the line between bus 3 and bus 4, and finally rerun the power
↳ flow, then list every line whose loading is above 80%.
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 4, "p_mw": 42.7}) -> disconnect_line({"from_bus": 3, "to_bus":
↳ 4}) -> reconnect_line({"from_bus": 3, "to_bus": 4}) -> run_powerflow({})
executed_calls: load_case({"case_name": "case14"}) -> get_status({}) -> modify_load({"bus_id": 4, "p_mw": 42.7, "q_mvar": -3.9}) ->
↳ disconnect_line({"from_bus": 3, "to_bus": 4}) -> reconnect_line({"from_bus": 3, "to_bus": 4}) -> run_powerflow({})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 251.83093927460206,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: get_status({})
output: {"active_case": "case14", "network_info": {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11,
↳ "total_load_mw": 251.83093927460206, "total_gen_capacity_mw": 772.4000000000001}, "has_last_result": false}
gate: n/a (no es un resultado de flujo)
--- Ronda 3 ---
tool: modify_load({"bus_id": 4, "p_mw": 42.7, "q_mvar": -3.9})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -4.656433439016519, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.009999 ...[6330 chars más]
gate: PASS (reasons=-, mismatch=4.73e-09 MW)
--- Ronda 4 ---
tool: disconnect_line({"from_bus": 3, "to_bus": 4})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.8175291024652545, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.009999 ...[6330 chars más]
gate: PASS (reasons=-, mismatch=2.35e-09 MW)
--- Ronda 5 ---
tool: reconnect_line({"from_bus": 3, "to_bus": 4})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -4.656433439016519, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.009999 ...[6330 chars más]
gate: PASS (reasons=-, mismatch=4.73e-09 MW)
```

```

--- Ronda 6 ---
tool: run_powerflow({})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -4.656433439016519, "is_violation": false,
  ↪ "violation_type": null}, {"bus_id": 3, "vm_p ...[6396 chars más]
gate: PASS (reasons=, mismatch=4.73e-09 MW)
--- Ronda 7 ---
LLM: The requested sequence completed successfully, and the final AC power flow converged with the 3-4 line reconnected and bus 4 active
  ↪ load set to 42.7 MW.

**Key numbers**
- Total load: **247.749 MW**
- Total generation: **259.607 MW**
- Total losses: **11.858 MW**
- Voltage range: **1.010-1.090 p.u.* ...[347 chars más]
--- Respuesta final ---
The requested sequence completed successfully, and the final AC power flow converged with the 3-4 line reconnected and bus 4 active load
  ↪ set to 42.7 MW.

**Key numbers**
- Total load: **247.749 MW**
- Total generation: **259.607 MW**
- Total losses: **11.858 MW**
- Voltage range: **1.010-1.090 p.u.**

**Violations**
- **Lines loaded above 80%: None**
- No thermal violations were reported.
- Overvoltage violations occurred at buses **1, 6, 7, 8, 9, 10, 11, 12, and 13**.
- Highest voltage: **1.090 p.u. at bus 8**

**Suggested next step:** Review generator voltage setpoints or run remedial-action analysis to reduce the overvoltage violations.
--- Scoring ---
formulation=extra_arg_changes_result (exact=False); V_MAE=8.15e-06; F_MAE=0.00; faithful=100.0 %; solved=False (formulation); flags=-;
  ↪ error=-; tokens=35535; cost_usd=0.0786; wall=38.0 s

```

Ejemplo: con bandera de fallo (claimed/stale/abstained)

```

request (case14-plain-004-s0, case14, difficulty=plain, seed=903170603): Load case14 and run an N-1 contingency analysis and report the
  ↪ worst single outage.
intended_calls: load_case({"case_name": "case14"}) -> run_n1_contingency({})
executed_calls: load_case({"case_name": "case14"}) -> run_n1_contingency({"top_k": 1, "criteria": "max_violations", "max_candidates": 0})
  ↪ -> get_status({}) -> disconnect_line({"from_bus": 1, "to_bus": 5}) -> reconnect_line({"from_bus": 1, "to_bus": 5})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 261.9932169620855,
  ↪ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: run_n1_contingency({"top_k": 1, "criteria": "max_violations", "max_candidates": 0})
output: {"plot_type": "n1_ranking", "figure_json":
  ↪ "{ \"data\": [{ \"cliponaxis\": false, \"hoverinfo\": \"text\", \"hovertext\": [ \"\\u003cb\\u003eOutage\\u003c\\u002fb\\u003e: 1-5
  ↪ (line)\\u003cbr\\u003e\\u003c\\u003eConverged\\u003c\\u002fb\\u003e: True\\u003cbr\\u003e\\u003cb\\u003eVoltage
  ↪ violations\\u003c\\u003e ...[9156 chars más]
gate: n/a (no es un resultado de flujo)
--- Ronda 3 ---
tool: get_status({})
output: {"active_case": "case14", "network_info": {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11,
  ↪ "total_load_mw": 261.9932169620855, "total_gen_capacity_mw": 772.4000000000001, "has_last_result": false}
gate: n/a (no es un resultado de flujo)
--- Ronda 4 ---
tool: disconnect_line({"from_bus": 1, "to_bus": 5})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000004, "va_deg": -7.875088276323269, "is_violation": false,
  ↪ "violation_type": null}, {"bus_id": 3, "vm_p ...[6329 chars más]
gate: PASS (reasons=, mismatch=2.03e-08 MW)
--- Ronda 5 ---
tool: reconnect_line({"from_bus": 1, "to_bus": 5})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -5.12793878696546, "is_violation": false,
  ↪ "violation_type": null}, {"bus_id": 3, "vm_pu ...[6435 chars más]
gate: PASS (reasons=, mismatch=4.7e-09 MW)
--- Ronda 6 ---
LLM: The N-1 analysis of all 20 branches converged and ranked the **line 1-5 outage** as the worst single contingency under the
  ↪ **maximum-violations** criterion.

**Key numbers for the worst outage**
- Total load: **261.993 MW**
- Total generation: **283.572 MW**
- Total active-power losses: **21.579 MW* ...[478 chars más]
--- Respuesta final ---
The N-1 analysis of all 20 branches converged and ranked the **line 1-5 outage** as the worst single contingency under the
  ↪ **maximum-violations** criterion.

**Key numbers for the worst outage**
- Total load: **261.993 MW**

```

```

- Total generation: **283.572 MW**
- Total active-power losses: **21.579 MW**
- Voltage range: **1.0075-1.0900 pu**
- Maximum branch loading: **2.362%**

**Violations**
- **9 overvoltage violations**
- **0 thermal violations**
- Change from base case: **0 additional voltage violations, 0 additional thermal violations**
- The power flow converged successfully.

**Suggested next step:** Run the ranking by maximum overload or minimum voltage to assess severity from alternative criteria; the 1-5 line
↪ has been reconnected and the base topology restored.
--- Scoring ---
formulation=extra_step (exact=False); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=False (formulation); flags=stale_state; error=-;
↪ tokens=28827; cost_usd=0.0634; wall=39.0 s

```

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- Todos los métodos que llegan al solver (react_nogate) tienen $V_MAE < 1e-4$ p.u. en todos los casos (máximo 8.15e-07 en react_nogate/case14).
- El gate verificó 16 resultados de flujo, 0 fallaron la verificación y se activó (retuvo números) 0 veces en 0 items.
- Ningún item terminó con **error**.
- Solved de **react_nogate**: 70.0 % (7/10).
- Formulación exacta agregada sobre métodos con tools: 70.0 % (7/10).
- Formulación por dificultad: mejor **parameterized** 100.0 % (3/3), peor **plain** 33.3 % (1/3).
- Errores de formulación más frecuentes: **extra_arg_changes_result=2**, **extra_step=1**.
- Ningún item reclamó éxito sobre un fallo (claimed_success_on_failure=0).
- 1 de 7 items con mutación de red reportaron estado obsoleto (stale_state).
- Trazabilidad de números (Faith.) media con tools: 100.0 %.
- Costo total 0.3268 USD; el método más caro fue **react_nogate** con 0.3268 USD (100 %).
- Métodos con tools: 3.20 tool calls medias por item (máximo 6).

6. Archivos

Rutas absolutas bajo **react_nogate.report.rescored.json** (cuando existe) es el archivo leído para las métricas.

```

report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↪ /power-flow-agent/benchmarks/results_validation_n10/gpt-5.6-sol/react_nogate/report.json
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↪ /power-flow-agent/benchmarks/results_validation_n10/gpt-5.6-sol/react_nogate/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↪ /power-flow-agent/benchmarks/results_validation_n10/gpt-5.6-sol/react_nogate/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↪ /power-flow-agent/benchmarks/results_validation_n10/gpt-5.6-sol/react_nogate/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↪ /power-flow-agent/benchmarks/results_validation_n10/gpt-5.6-sol/react_nogate/scoreboard_per_case.json
traces (10 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↪ /power-flow-agent/benchmarks/results_validation_n10/gpt-5.6-sol/react_nogate/traces

```